Report: Order Delivery Time Prediction

Objectives

The objective of this assignment is to build a regression model that predicts the delivery time for orders placed through Porter. The model will use various features such as the items ordered, the restaurant location, the order protocol, and the availability of delivery partners.

The key goals are:

- Predict the delivery time for an order based on multiple input features
- Improve delivery time predictions to optimise operational efficiency
- Understand the key factors influencing delivery time to enhance the model's accuracy

Below are the Steps taken:

1. Data Preparation:

- Fixing the Datatypes : Converted actual_delivery_time and created_at to datetime format using pandas
- Also converted Categorical fields to type Category.

2. Feature Engineering:

- Made the predictor variable time_taken in minutes by subtracting the date columns, and using pd.Timedelta(minutes=1), converted it into minutes.
- Extracted pickup_hour from created_at and day_of_week variable to extract another derived variable isWeekend by checking if day of week is Saturday or Sunday
- Then I removed the date columns and day_of_week to avoid redundancy.

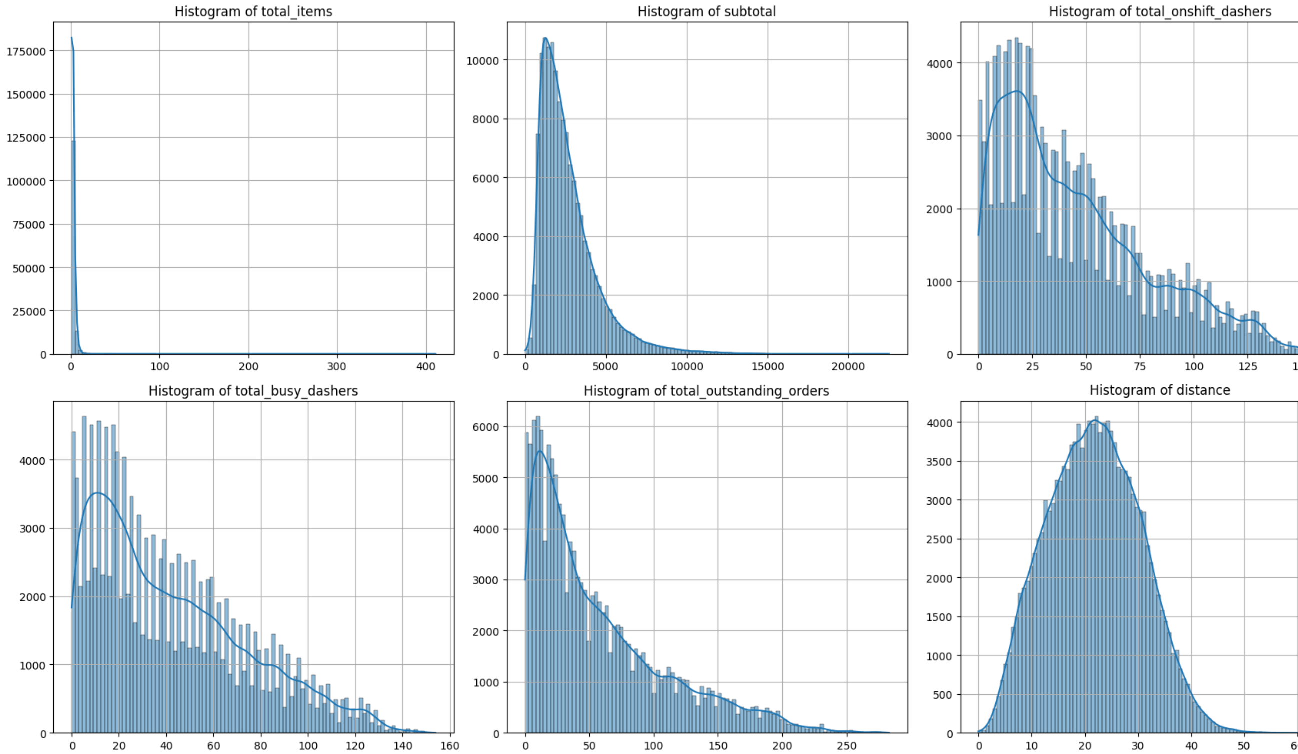
3. Creating Training and Validation sets:

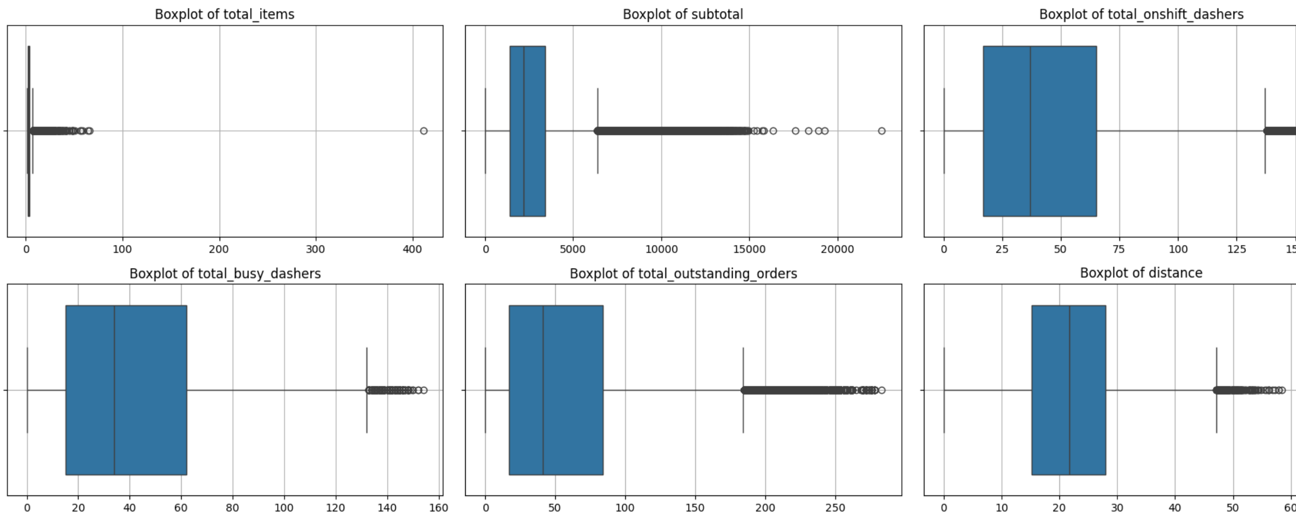
- The dataset was split into training and testing subsets using an 80-20 ratio. Further split into X_train, y_train, X_test, y_test. This step ensured that model performance could be evaluated on unseen data.

4. Exploratory Data Analysis on Training Data:

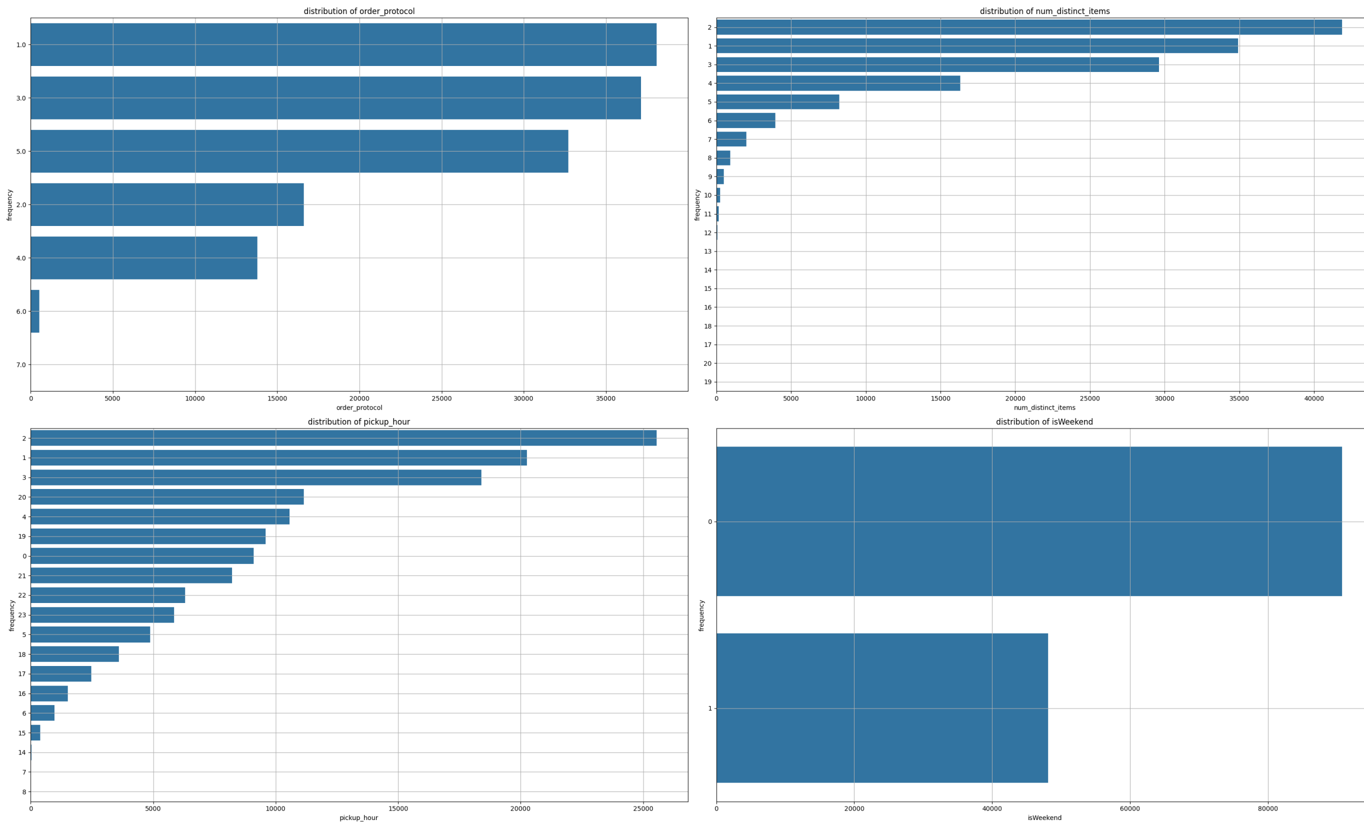
4.1. Visualizing the Distribution of Numerical Features:

After plotting the histograms and box plots of numerical columns, I was able to see the skewness in data and potential outliers in these columns.

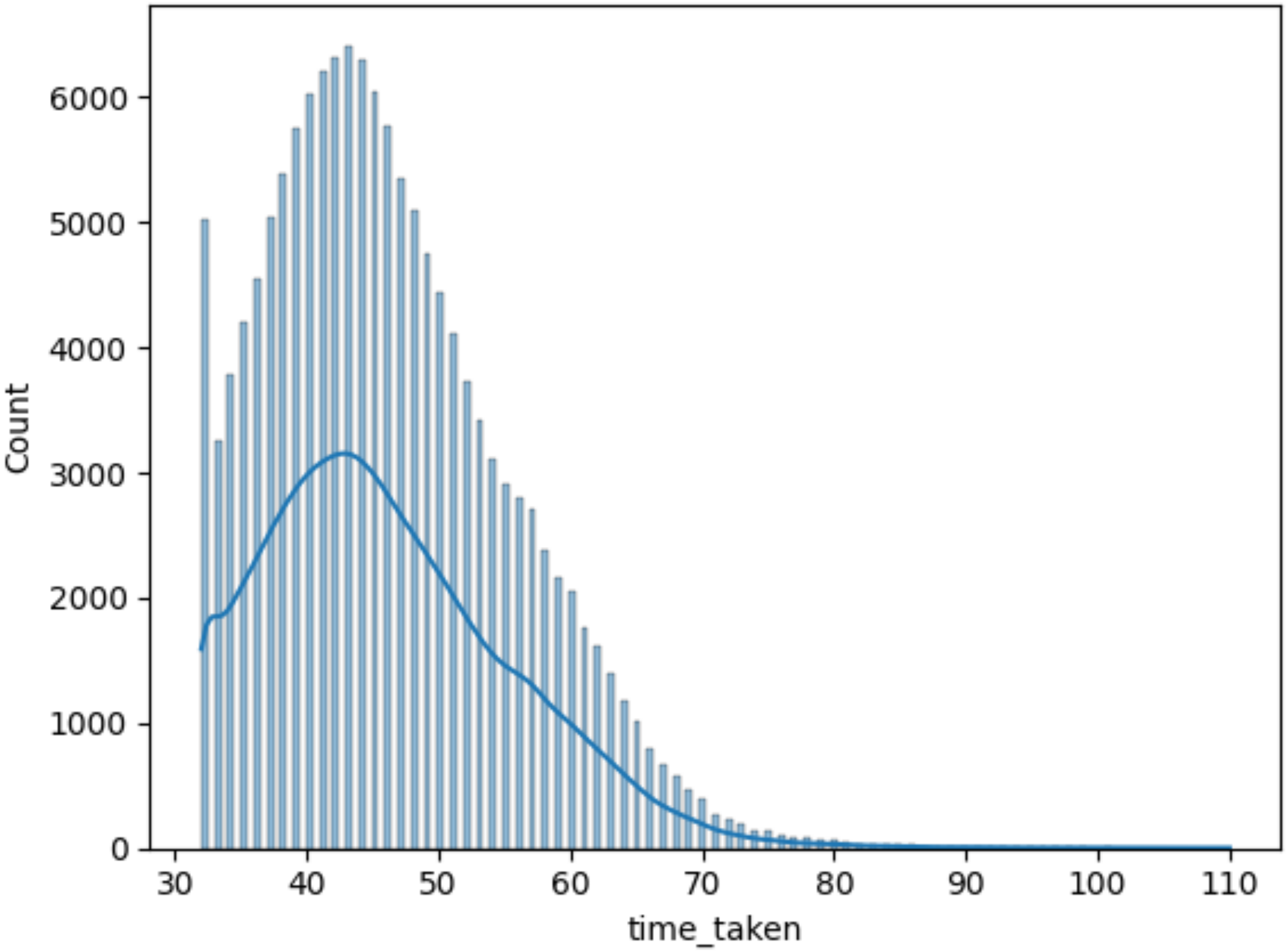




For categorical variables, I found this trend.

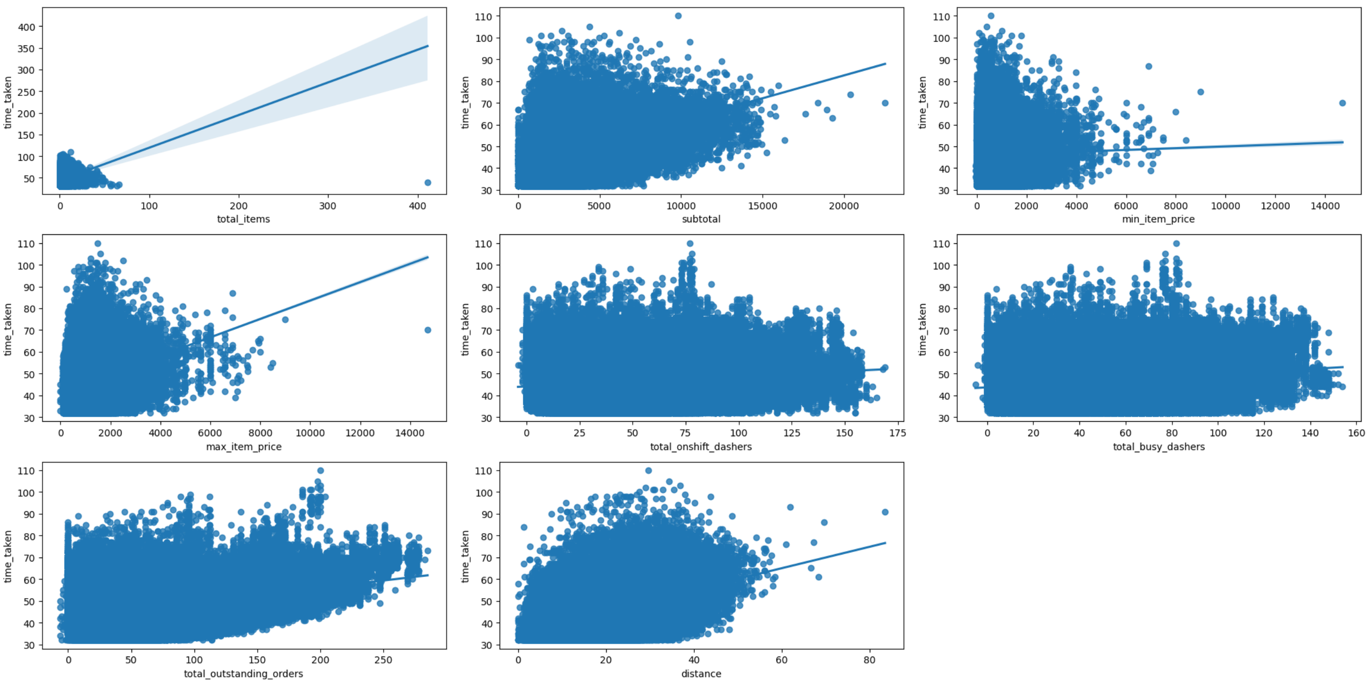


4.2 Distribution of Time Taken (Target Variable):



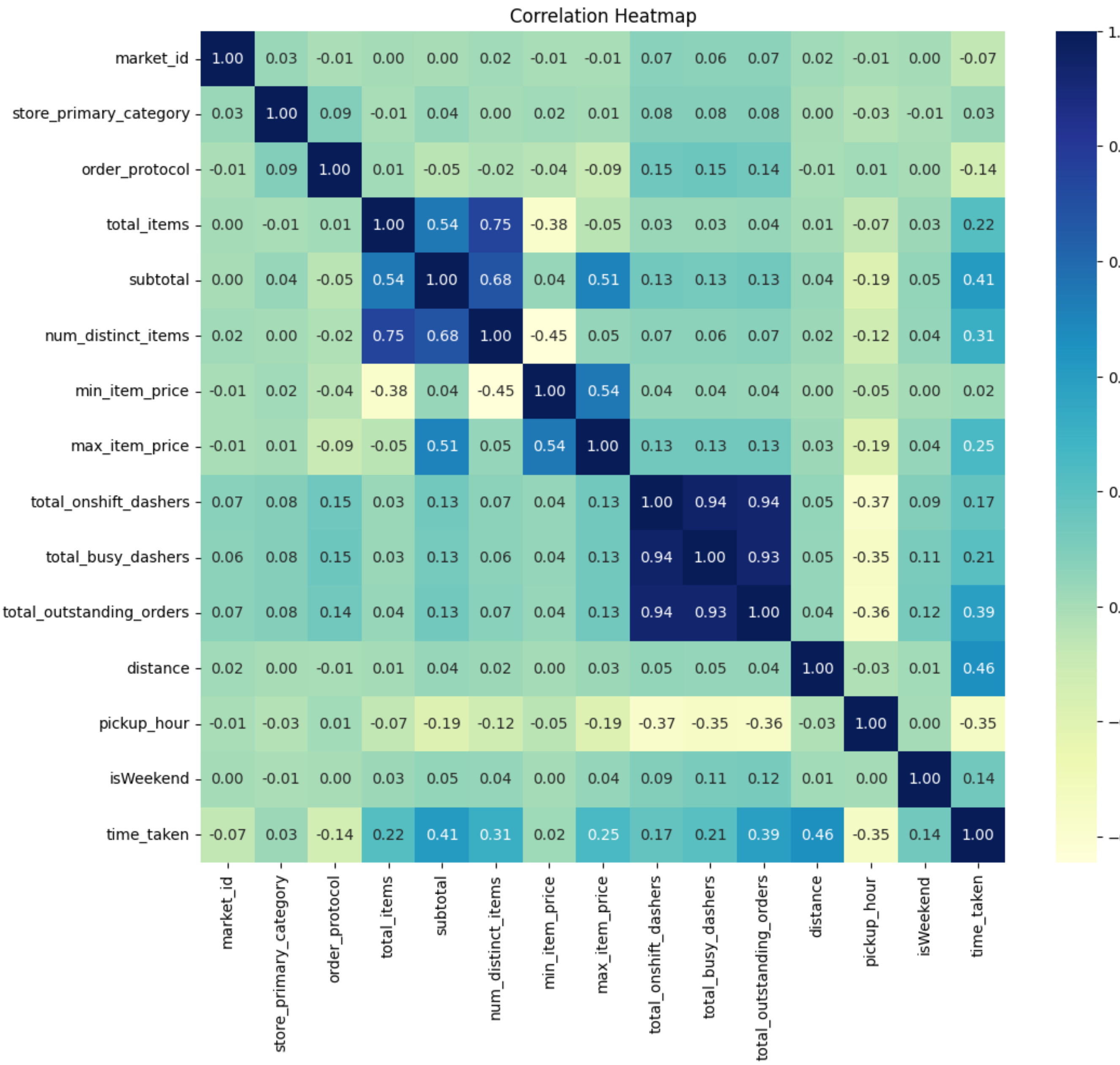
From the image, we can see that the predictor variable is right skewed.

The relation between numerical features and dependent variable is shown below:


time_taken
total_items
subtotal
min_item_price
max_item_price
total_onshift_dashers
total_busy_dashers
total_outstanding_orders
distance

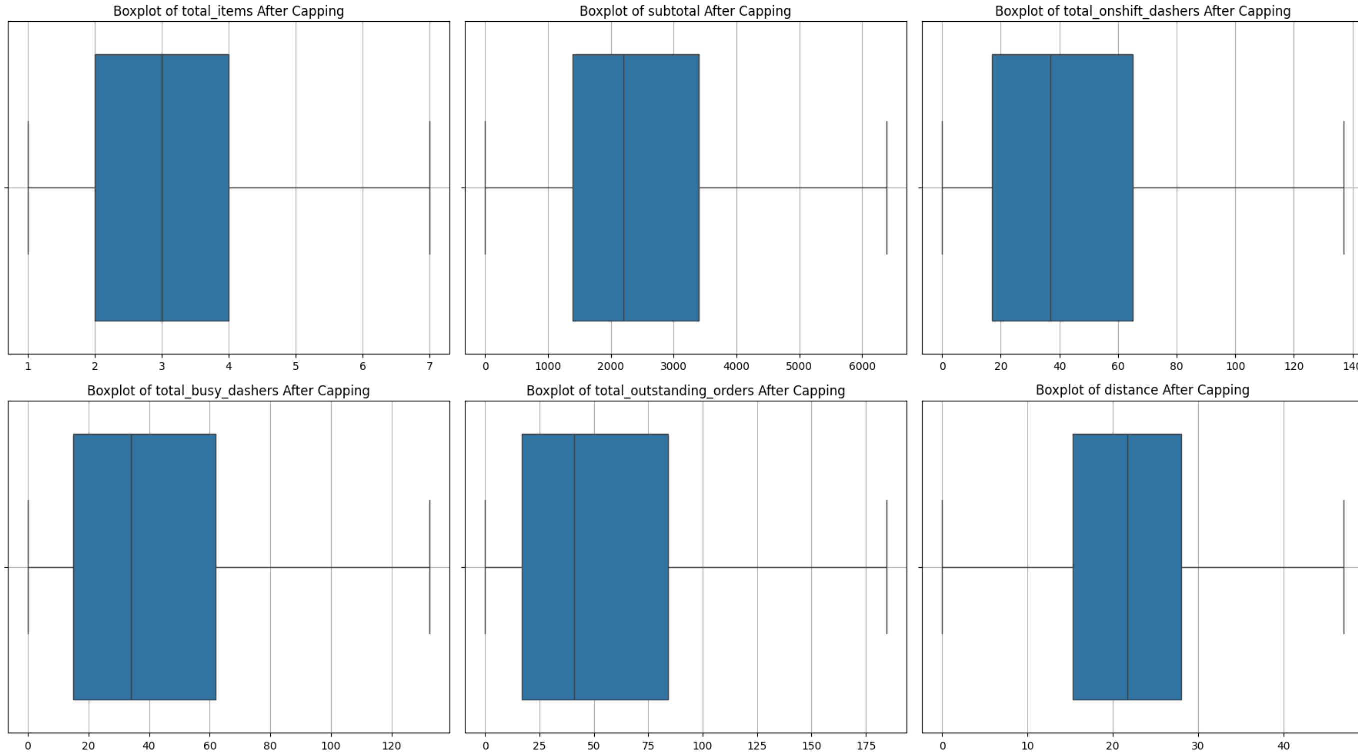
4.3 Correlation Analysis:

4.3.1 After plotting the heatmap, I was able to conclude that distance had the highest correlation with the target variable and I dropped 3 features having weak correlation with the target variable: store_primary_category,market_id and min_item_price.



5. Handling Outliers:

5.1.1 Visualized and calculated the number of outliers in each column. My strategy of handling outliers was to cap some outliers and remove extreme outliers. This ensured minimal loss of data and I was hoping to get more accurate results. Here are the plots after capping and removing outliers



6. Model Building:

1. Feature Scaling:

1.1. MinMaxScaler is used to scale the numerical features. The MinMaxScaler transforms each feature so that it falls within a specified range (typically between 0 and 1). I transformed the train data using the function fit_transform, and the test data using transform after keeping the same columns in the test dataset as in the train data set

2. Steps For building the Linear Regression Model:

2.1. I used scikit-learn's Linear Regression module, and after fitting it to the test and train data, I predicted both the train data and test data to compare the metrics.

2.2. I built a function to calculate all the evaluation metrics including, r2-score,Mean squared Error, Mean Absolute Error and Adjusted R Sqaured.

2.3. These were my findings:

Statistic	Value
R^2	0.8656
Adjusted R^2	0.8656
MAE	2.4540
MSE	11.5404
RMSE	3.3971

2.4 Then I used stats models library to find additional details about the model

Findings:

a) All p values were 0, implicitly meaning all were significant

b) Probability of F statistic was zero, which showed the model was significant

c) But the statistic showed there was multi collinearity in my data as the conditioning number was high

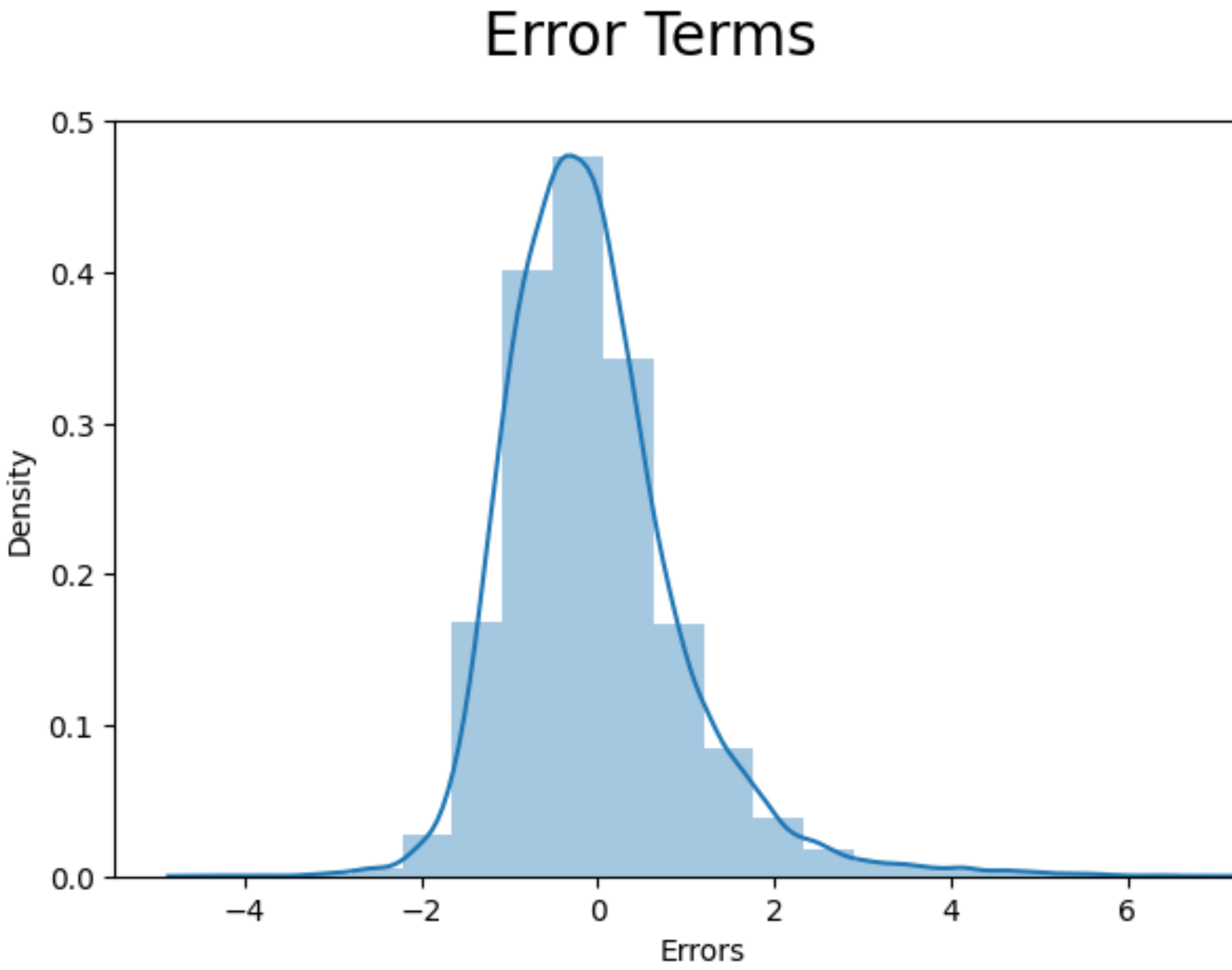
2.5 Then I calculated the variance inflation factor, and found some features to be highly correlated

7. Build the model and fit RFE to select the most important features [7 marks]:

I iteratively built the linear regression model using Recursive Feature Elimination, checking summary statistics at each step and finally found a model containing 8 features… which had minimal correlation and were highly significant.
RFE Model Evaluation Statistics:
R^2: 0.8148
Adjusted R^2: 0.8147
MAE: 2.9708
MSE: 15.9008
RMSE: 3.9876
Then using the stats model library, I calculated the detailed statistics and variance inflation factor.

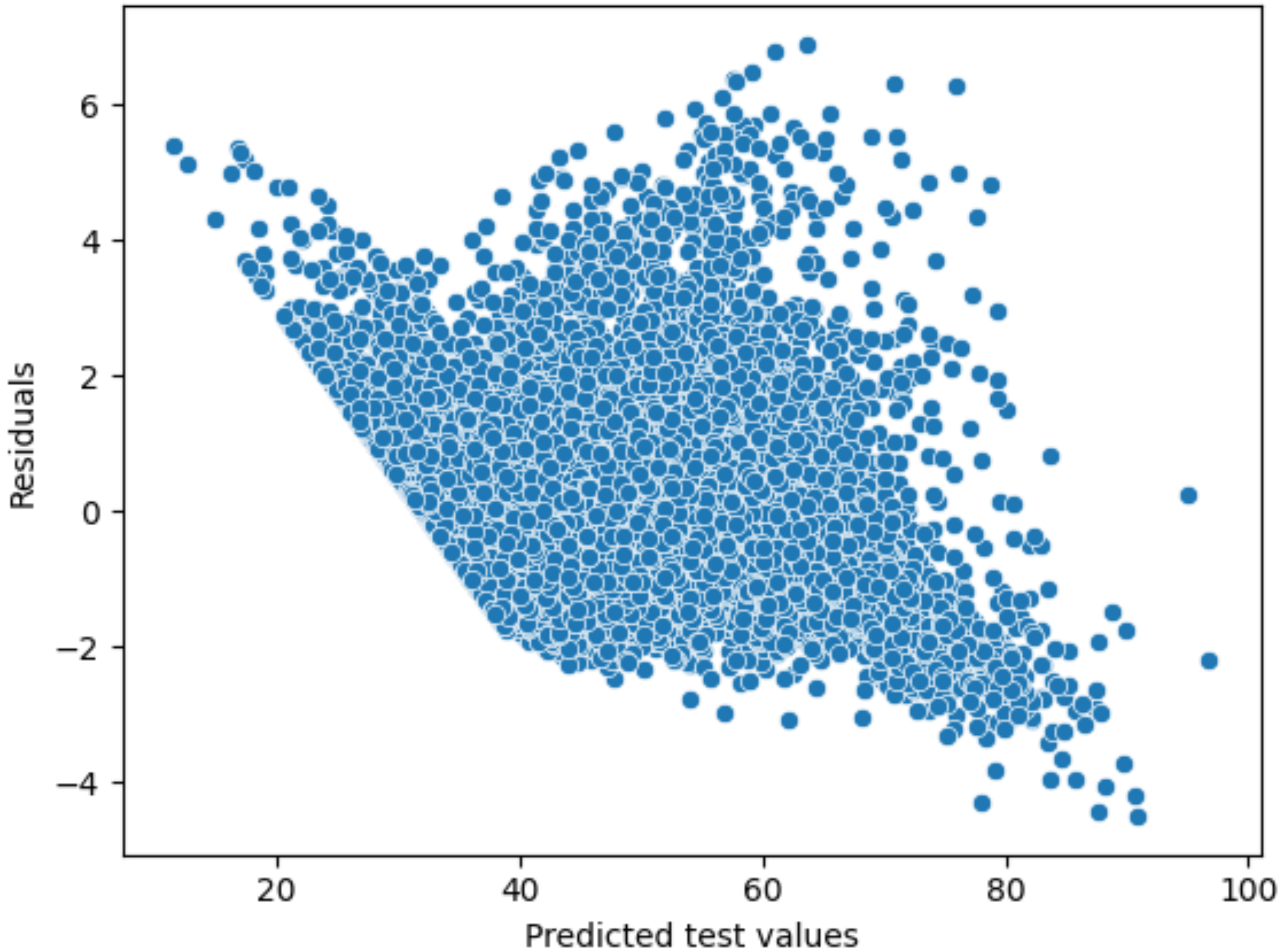
8. Residual Analysis: Residual plots help us check if the assumptions of linear regression hold for our model.

8.1. **Histogram of Residuals**: I found the distribution of error terms approximately close to normal as shown

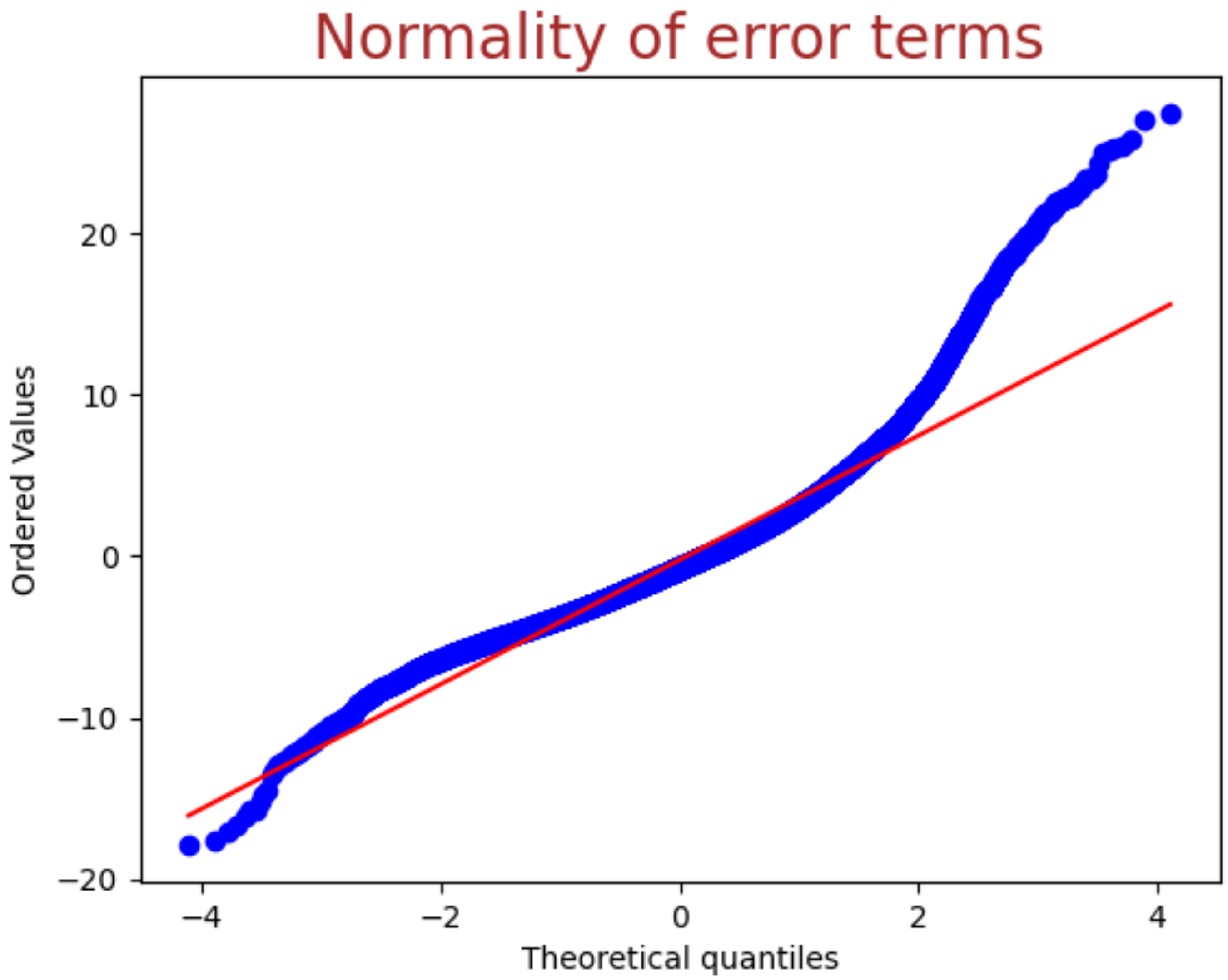


8.2 Checking for heteroscedasticity by plotting predicted vs residuals. There did not seem to be much relation as shown.

Residuals vs predicted



8.2. **Finally I built the quantile quantile plot of residuals as shown:**



9. Coefficient Analysis: Using the unscaled data, I built the linear regression model again and found the coefficients had much lower magnitude, than when I had transformed the data.
   9.1. Also I found 1 unit change of total items, had slight change in the coeffients.

7. Subjective Questions were answered in assignment.